AIR POLLUTION DETECTION

Manvi Bamal E23BCAU0137

Shambhavi Kumari E23BCAU0134

Acknowledgment

I would like to take this opportunity to sincerely thank everyone who supported me throughout the duration of this project. I express my profound gratitude to my project supervisor for their constant guidance, timely advice, and encouragement. Their insights greatly contributed to the successful completion of this project. I would also like to thank my faculty members for their invaluable inputs during discussions, and my teammates for their cooperation, commitment, and hard work. Without their continuous motivation and contribution, it would have been difficult to bring this project to its current form. Lastly, I am thankful to my friends and family for their moral support throughout this journey.

Abstract

QuickFix Roadside Service is an innovative mobile application aimed at revolutionizing roadside assistance for travellers. The app provides a comprehensive network of real-time services including emergency support, access to nearby garages and fuel stations, live location sharing for enhanced safety, and multilingual options for ease of use across different regions. By integrating various service providers into a single, user-friendly platform, the application seeks to solve critical problems faced by travellers, especially during emergencies on highways and remote roads. The target audience includes regular travellers, solo adventurers, women travellers, electric vehicle users, and international tourists who may face language barriers. Through this project, we aim to ensure that help is always just a click away for anyone on the road.

Table of Contents

5. Introduction

In today’s fast-moving world, traveling between cities has become a common part of life, whether for work, leisure, or personal reasons. However, travellers often encounter unexpected difficulties such as car breakdowns, fuel shortages, or medical emergencies in unfamiliar locations. Unfortunately, reliable assistance is not always readily available, especially in remote areas. This lack of real-time help can cause stress, delay, and even pose serious safety risks.

The QuickFix Roadside Service application was conceptualized to address these exact challenges. The project's main objective is to develop a mobile platform that provides travellers with instant access to roadside assistance services based on their real-time location. By connecting users with verified service providers and offering emergency features like live location sharing and an SOS button, the application aims to enhance road safety and convenience. The primary users include road trippers, solo and women travellers, tourists facing language barriers, and owners of electric vehicles who require access to nearby EV charging stations. In a rapidly evolving digital era, this project promises to contribute meaningfully to smarter, safer travel experiences.

6. Tools & Technologies Used

The development of QuickFix Roadside Service involved the careful selection of modern, reliable, and scalable tools and technologies. For the frontend development, we utilized React Native, which enabled us to create a cross-platform mobile application compatible with both Android and iOS devices. React Native provided flexibility and efficiency, reducing development time and ensuring consistent performance across platforms.

For backend development, we employed Node.js and Express.js to handle server-side logic, authentication, and database management. Firebase was chosen as the cloud platform for its real-time database services and secure authentication capabilities, essential for emergency features like live tracking. Mapping and location-based services were integrated using the Google Maps API, which allowed real-time access to nearby garages, fuel stations, and EV charging points. In addition, the Google Translate API was used to implement multilingual support, breaking language barriers for international users. For hosting and deployment purposes, Firebase Hosting was considered the best option due to its integration with our tech stack and reliable uptime.

7. System Architecture / Design

The architecture of QuickFix Roadside Service was carefully designed to ensure reliability, scalability, and a seamless user experience. At the core, the application is structured around location-based service delivery. The app’s sitemap includes primary sections such as Home, Emergency Help, Nearby Services, Live Tracking, and Profile Settings.

Wireframes were initially created using Figma, allowing us to visualize user journeys and refine the interface design for simplicity and clarity. UI/UX decisions prioritized ease of navigation, quick access to emergency features like the SOS button, and a clean visual hierarchy to avoid confusion during stressful situations. Special emphasis was placed on ensuring the app was friendly to solo and women travellers through features like trusted service providers and secure live location sharing. On the backend, Firebase Cloud Firestore was chosen to manage user profiles, service provider listings, and real-time updates. The design allowed for future scalability, including adding features like traffic updates and health emergency integrations.

---

8. Implementation / Development

The implementation phase involved rigorous planning and coordination among the team members to bring the envisioned application to life. The project started with developing wireframes and finalizing the list of core features, ensuring that the application would meet the identified needs.

Initially, the backend setup was completed, including the server configurations and database structure. APIs such as Google Maps and EV Charging Station APIs were integrated to fetch live data. Following the backend setup, frontend development began with the creation of screens such as Home, Nearby Services, Emergency Help, and Profile Management. Each screen was designed with user-centred principles, ensuring intuitive navigation and quick access to critical functions like the SOS button.

Live location sharing and multilingual functionality were developed to ensure the application could cater to a global audience. Real-time data synchronization between frontend and backend was a major focus, ensuring that users received accurate and updated information at all times. Screenshots of different stages of app development were maintained as evidence of progressive achievement.

9. Testing

Testing was a critical phase in ensuring the application's reliability and performance under various real-world conditions. A combination of manual testing and functional testing was employed across multiple Android and iOS devices. Core functionalities such as the SOS button, location sharing, and nearby service listing were tested extensively to ensure stability and responsiveness.

We conducted browser and device compatibility checks to guarantee that users on different platforms experienced no disruption. During the testing phase, several bugs related to location fetching under low network conditions and UI glitches on smaller screens were identified and rectified. The testing phase also included stress testing emergency features to measure their responsiveness under network fluctuations. Final user acceptance testing (UAT) was conducted with a small group of beta testers, whose feedback helped further refine the user interface and feature performance.

10. Challenges Faced

Every project faces its set of challenges, and QuickFix Roadside Service was no exception. One of the significant technical hurdles was the integration of multiple APIs for location, fuel prices, and EV charging station data, all while maintaining real-time performance. Ensuring multilingual support dynamically was another complex task, as translation errors needed to be minimized to avoid miscommunication during emergencies.

Live location sharing also posed challenges, particularly when dealing with network instability in remote areas. Balancing feature richness with application performance and battery optimization required several iterations of code refactoring. Additionally, coordinating among team members to manage frontend, backend, and testing activities simultaneously tested our time management and collaboration skills. Despite these obstacles, the team stayed motivated and successfully overcame each challenge through careful planning and consistent communication.

11. Results

The final output of the project was a fully functional prototype of the QuickFix Roadside Service mobile application, achieving all the major objectives set at the project's start. The application successfully integrated emergency features, real-time service listings, fuel price comparison, multilingual support, and user reviews, creating a comprehensive solution for travellers.

Performance testing results showed stable operation across various network conditions and devices. The app's emergency features, such as the SOS button and live location sharing, performed reliably, providing users with a sense of security. Initial beta testing feedback was overwhelmingly positive, with users particularly appreciating the safety features and multilingual interface. Based on performance metrics and feedback received, the project can be considered a significant success, with strong potential for future expansion and public launch.

12. Conclusion

Through the QuickFix Roadside Service project, we gained invaluable practical experience in mobile application development, system integration, user-centered design, and real-time service delivery. The project allowed us to solve a real-world problem using modern technological solutions, enhancing our technical skills as well as our project management capabilities.

All the primary objectives were achieved within the planned timeline, and the app successfully demonstrated the feasibility and importance of providing real-time roadside assistance to travellers. Looking ahead, future improvements could include AI-driven suggestions, offline mode functionality, and deeper integration with emergency services and traffic management systems. The knowledge and experience gained from this project will undoubtedly be instrumental in our future endeavours in the tech industry.

---

13. References

- Adams, M.D., & Jones, T.W.W. (2018). Mobile Application Development with Web Technologies.

- Poushneh, A. (2018). The Role of Mobile Location-Based Services in Consumer Behavior - Journal of Mobile Marketing.

- Hwang, J., & Kim, H. (2017). User Preferences for Location-Based Mobile Services - International Journal of Human-Computer Interaction.

- Patel, S., et al. (2016). Mobile Applications for Emergency Response - International Journal of Emergency Management.

- Xu, H., & Zhang, L. (2018). Impact of Multilingual Support in Mobile Apps for International Travelers -